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Curcumin

Age-old botanical,
modern-day results



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Viewpoint: Science is bridging a wide chasm when it comes to curcumin

Modern clinical research is uncovering the popular ingredient's complex chemistry, notes **Duffy Hayes**, and affirming centuries of folk medicine in a compelling way.



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Innovation and versatility play a role in curcumin's future growth

Innovative delivery methods, new research regarding the best absorption methods, and curcumin's long-standing history all lend themselves to its increased popularity, writes **Joy Choquette**.



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Research continues to be a critical tool unlocking the potential of curcumin, explains **Rachel Adams**, allowing natural products manufacturers, marketers and consumers to seize the power of this super compound.



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Curcumin in 2021: Some answers, but questions remain

A wealth of studies were published this year by specific ingredient backers aiming to provide answers about absorption and bioavailability, but there's been little clarity as a result, asserts **Steve Myers**.



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Curcumin: Keeping abuzz with integrity, innovation

Pete Croatto describes the challenges associated with the commoditization and source redundancy happening around curcumin, as the ingredient's outsized demand meets at a nexus with its sky-high potential.



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Science is bridging a wide chasm when it comes to curcumin



This is an exciting time in the world of supplements. What is happening with curcumin—where current clinical research is uncovering the complex chemistry behind the compound's well-known mechanisms of action, previously not elucidated in folk medicine—is particularly emblematic.

As with many historically tried-and-true remedies, the gap between the utility of a substance and our knowledge of how it actually affects the human system is narrowing, as an emphasis on the use of natural and complementary approaches in Western medicine draws the attention of researchers.

Curcumin—the nutritional compound located within the rhizome, or rootstalk, of the turmeric (*Curcuma longa*) plant—continues to play a leading role in India's ancient medical system of ayurveda. And the range of uses in the current version of the practice is fairly astonishing.

Medical dressings made of turmeric paste still are used to treat common eye infections, and to dress wounds, treat bites, burns, acne and various skin diseases. Powdered turmeric is taken with boiled milk to cure cough and related respiratory ailments, and roasted turmeric is an ingredient used as a medicine against dysentery in children, according to the book “[Major medicinal plants of India](#),” published by the Central Institute of Medicinal and Aromatic Plants in Lucknow, India.

In some parts of India, women are given a tonic of fresh turmeric paste with powder of dried ginger roots and honey in a glass of hot milk to drink twice daily after childbirth. Dressings of turmeric paste are also applied to the perineum to help heal lacerations in the birth canal (Trees Life J. 2005;1).

More wide-ranging uses of turmeric in ayurveda include applications in dentistry and digestion. In another creative if not fully understood mechanism of action, it's shown to alleviate the hallucinatory effects of hashish and other psychotropic drugs (Phytother Res. 2004;18:798-804).

While the issue that unfolds on the following pages won't fully explain the functionality behind all of these folk remedies and uses, it does lay out with specificity some of the strides science is making to uncover the mysteries of curcumin's history, especially related to gut dyspepsia, joint and muscle protection, and cognitive function.

Balanced against the functional as well as market challenges that are further described, this issue hopefully serves as a small journey of discovery, mixed with some real-world supplement business application. Enjoy this modest dose of enlightenment!

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The benefits and potential of curcumin

by Rachel Adams



Curcumin is a chemical compound isolated from *Curcuma longa* L. (turmeric) plants that is known in the nutrition community for its potential anti-inflammatory, anticancer and antimicrobial properties.¹

With a global market share of more than US\$58 million in 2020, and a projected compound annual growth rate (CAGR) of more than 16% from 2020 to 2028—according to a [report](#) from Grand View Research—this compound is a shooting star in the natural products industry that continues to draw attention from researchers and consumers alike.

Fueling the growth of curcumin is clinical research, which connects consumers to the health benefits of this nutritional player. Emerging studies are extending the reach of curcumin, opening doors to new health avenues that bring opportunities for natural products marketers and manufacturers, as well as the customers who take stock in their products.

A healthy gut

Functional dyspepsia is a condition characterized by recurring symptoms of indigestion, such as upper abdomen discomfort, bloating, belching and nausea. According to N. Kalyanam, Ph.D., president of research and development (R&D) at Sabinsa Corp., the ailment can also be compounded with *H. pylori* infection. “Functional dyspepsia and gut-related problems affect a good fraction of the population and lower the quality of life, especially due to their persistence or recurrence,” he explained.

Research this year suggests curcumin may provide support for those dealing with functional dyspepsia. In a one month, triple-blinded clinical study of 75 subjects with functional dyspepsia, 500 mg daily standardized turmeric extract with 95% curcuminoids (as Curcumin C3 Complex with Bioperine, from Sabinsa) as an adjuvant to a standard therapy decreased various indicators of upper abdominal discomfort in a statistically significant fashion, as compared to the group receiving only standard therapy.² The positive outcomes were recorded upon completion of the treatment, and during follow-up a month later. Results included a higher eradication rate of *H. pylori* in the curcumin group as well.

Curcumin further supported the gut in a double-blind, placebo-controlled, randomized digestive health study of 80 subjects with self-reported digestive and mood complaints.³ Participants were administered either a turmeric-oleoresin extract with 50% curcuminoids (500 mg as Curcugen, from DoiCas Biotech) or placebo for eight weeks. Changes to abdominal pain, reflux, diarrhea, constipation and other symptoms were monitored using a gastrointestinal symptom rating score (GSRS) inventory. The Curcugen group scored significantly better in the reduction of abdominal pain, compared to placebo (28% vs. 18%), with the placebo group getting worse during weeks four through eight.

What's more, researchers noted a significant reduction in anxiety scores in the Curcugen group (52%), versus 16% in the placebo group.

Curcumin may also support a healthy gut microbiome. In a double-blind, placebo-controlled study, an inter-university team in the U.S. examined the effects of placebo, turmeric powder and C3 Complex+Bioperine on the gut microbial population, finding curcuminoids played a “prebiotic-like” role in the promotion of healthy gut microbial population.⁴

Enduring cognition

“Since the onset of the global pandemic, consumers have more frequently sought out natural solutions to cognitive well-being, mood, emotional and mental health, and many other subcategories in the brain and cognitive health space,” explained Kristen Marshall, digital marketing manager, Verdure Sciences. “In addition to increased interest driving demand for cognitive and brain solutions, there is increased demand from an aging population looking for natural solutions in a proactive approach to their health.”

Curcumin is answering the call, offering researched benefits to mood, memory, fatigue and overall cognitive function.

Supporting curcumin's role in cognitive preservation and function is a new study showing a *C. longa* extract with >90% curcuminoids (as HydroCurc, from Pharmako Biotechnologies) supplemented with iron increased serum brain-derived neurotrophic factor (BDNF), a protein found in the brain and key for maintenance of normal neuronal function and energy homeostasis.⁵ BDNF is also suggested to impact cognitive function, including learning and memory, according to Eric Meppem, commercial director, and George Kokkinis, technical director, Pharmako Biotechnologies.

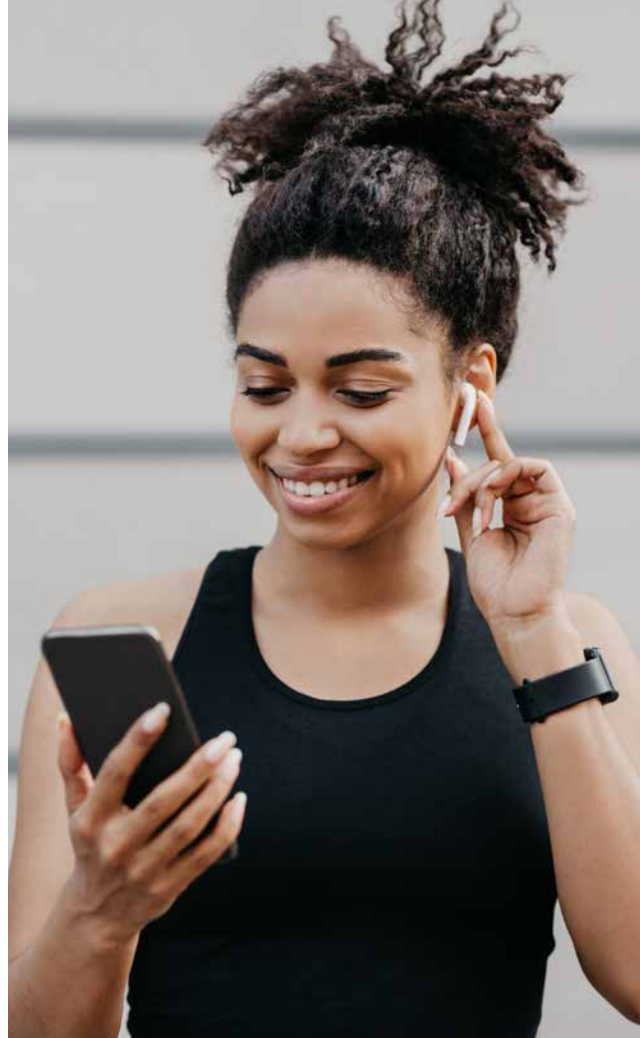
In the six-week study of 155 participants, HydroCurc co-administered with iron demonstrated a 26% increase in BDNF in the low-dose iron arm (18 mg) and a 35% increase in BDNF in the high-dose iron arm (65 mg).

“There is significant impact for both people concerned with cognitive decline, and people looking to improve cognitive performance (for example students, executives and e-gamers),” the duo said of the results.

One study found 400 mg (80 mg free curcumin) of solid lipid curcumin particle extract (as Longvida, from Verdure Sciences) improved working memory, sustained attention, spatial memory and learning, and mood in healthy, older adults over a 28-day period.⁶ A later partial replication study sought to extend the findings of the earlier trial by including a 12-week assessment in addition to a 28-day testing session.⁷

“The primary objective of this new clinical trial was to evaluate whether the effects of the four-week trial could be replicated following 12 weeks supplementation,” Marshall explained.

Findings showed curcumin (as Longvida at 400 mg/d) improved measures of mood and cognition, such as decreased fatigue, lowered tension and anxiety, improved working memory, sustained attention, and improved spatial memory and learning.



“Since the onset of the global pandemic, consumers have more frequently sought out natural solutions to cognitive well-being, mood, emotional and mental health, and many other subcategories in the brain and cognitive health space.”

Kristen Marshall, digital marketing manager, Verdure Sciences

“In conclusion, these data further support previous findings that Longvida curcumin improves working memory and mood as well as the possibility of learning in healthy individuals,” Marshall stated. “It is noteworthy that memory and fatigue are widely reported as the two more concerning non-physiological aspects of aging. This has potential promise to offset these effects and may also be relevant to conditions where mood and cognition are fragile.”

Add to that a 2021 study indicating Longvida crosses the blood-brain/retinal barriers,^{8,9} providing mechanistic data that is “further backed by clinical efficacy for a robust offering,” she said.

Athletic performance and recovery

For athletes, curcumin has the capacity to speed recovery and soothe sore muscles.

Kalyanam pointed to a 2021 study showing pre- and post-exercise curcuminoid supplementation attenuated muscle soreness while retaining muscle power in 22 adults.¹⁰ Participants were given 500 mg curcumin or placebo for 10 days: Six days pre-workout, the day of the workout and three days post-workout. Findings showed greater soreness in the placebo group compared to the curcumin group 48 hours and 72 hours post-exercise and a decrease in vertical jump over time in the placebo group, which was not seen in the curcumin group.

A 2021 double-blind, randomized clinical trial measured muscle pain following lower limb resistance exercise in 24 young, healthy males who were given placebo or curcumin dispersed in water (as HydroCurc) pre- and post-workout.¹¹ Results showed a significant reduction in delayed onset muscle soreness (DOMS) pain, positive activation of interleukin (IL)-6 and IL-10 inflammatory markers, reduced capillary lactate, and reduced thigh circumference at 24 hours and 48 hours in the HydroCurc group, compared to placebo.

Another study, presented during a 2020 meeting of the International Society of Sports Nutrition (ISSN), evaluated the potential anti-inflammatory and pain-relieving role of curcumin (a single 500 mg/d dose of Curcugen) in participants who completed intense exercise.¹² For the double-blind, placebo-controlled, randomized study, eccentric exercises created muscle damage in both groups, who were monitored over the next 72 hours. Findings showed a significant reduction in pain-to-pressure threshold (a measure of soreness) at hours 48 and 72, a reduction of high-sensitivity C-reactive protein (hsCRP, a marker of inflammation) at 72 hours, and a return to a normal range of motion at 72 hours in the Curcugen group, compared to placebo.

Adding to the growing body of evidence supporting curcumin for athletes is a 2017 randomized, controlled trial showing curcumin supplementation is effective in recovery following exercise-induced muscle damage.¹³ “The authors stated that the supplementation could attenuate muscle damage before it occurs,” Kalyanam reported.



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Youthful joints

Curcumin's benefits to joints are well documented. Despite an extensive repertoire of research, new studies evaluating curcumin's role in joint health are emerging.

In a new unpublished, randomized, double-blind, placebo-controlled study, researchers investigated the efficacy of curcumin (440 mg/d as HydroCurc) for alleviating joint pain and improving quality of life. Results showed a significant difference in participants' recorded pain value according to the visual analogue scale (VAS) over two weeks.

The impact of this study, according to Meppem and Kokkinis, derives from its demonstration of "the clinical benefits of HydroCurc at a one-a-day capsule dose, and over a relatively (14 days) short period of time."

A randomized, placebo-controlled clinical study explored the effects of 500 mg/d or 1,000 mg/d of a turmeric extract standardized to patented Turmerosaccharides (as Turmacin, from Natural Remedies) in 90 healthy volunteers feeling knee joint discomfort or pain of at least 40 mm on VAS during 10 minutes of a brisk walk on treadmill equipment.¹⁴ Findings of the 12-week study revealed Turmacin significantly decreased the final pain scores from the first follow-up of day five, compared to placebo. A maximum decrease of 1.7 units on the VAS scale from baseline was seen in the Turmacin group on days 56 and 84, a clinically significant difference. The product was also effective in improving joint range of motion and preserving muscle function during relative physical inactivity at the end of the study, compared to placebo.

Boosted immunity

Immune health has been at the forefront of the ongoing global health crisis. Curcumin's possible benefits to immunity are well-researched, with emerging evidence bolstering the compound's reputation.

In a 2021 double-blind, randomized, placebo-controlled trial of 40 healthy volunteers (60 years and older) 90 mg of amorphous solid dispersion of curcumin twice daily (as curcuRouge, from Robertet) for four weeks significantly decreased white blood cell count, neutrophil count and neutrophil ratio, and significantly increased lymphocyte ratio, resulting in a significant 11.3% decrease in neutrophil/lymphocyte ratio (NLR), compared to placebo.¹⁵

"Elevated NLR has been reported as a sensitive marker for predicting poor prognosis in chronic inflammation-based diseases such as stroke, heart failure, colorectal cancer and diabetes, as well as acute inflammatory diseases such as bacterial and viral infections," conveyed Laure Egoumenides, product manager at Robertet. "NLR is also known to increase with age and is considered to be an aging marker."

A key player affecting immune function is iron, with research showing a link between iron homeostasis and the immune system.¹⁶

Curcumin (as HydroCurc) was not only shown to have no negative impact on iron absorption in a 2021 publication, but the study showed the ability of curcumin to significantly elevate whole blood hemoglobin (Hgb) levels at 180 minutes compared to baseline in the group receiving the high-iron dose of 65 mg ferrous sulphate with HydroCurc.¹⁷

These findings, according to Meppem and Kokkinis, could encourage iron supplementation for those who need it. “The pro-oxidant effect of oral ferrous iron (the most cost-effective form) is well-known to induce inflammation, causing side effects,” the duo said, referencing the study publication. “These lead to poor compliance, with the end result of no health outcome improvements. This study demonstrates that those challenges can be overcome.”

Topical relief

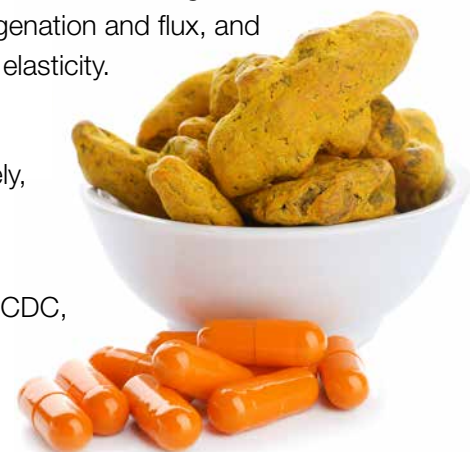
As curcumin research evolves, new areas such as skin health come into view. Atopic dermatitis, also known as eczema, is a condition characterized by red, itchy patches on the surface of the skin. [According to](#) the National Eczema Association, approximately 16.5 million U.S. adults have atopic dermatitis, with nearly 40% affected by moderate or severe disease. More than half (62.8%) of adults with moderate to severe disease report itching at least 12 hours per day.

Oral use of a phytosome curcuminoid complex (as Meriva, from Indena) in subjects experiencing atopic dermatitis and treated with moisturizers and ceramide creams impacted skin reactions to internal and external stimulations.¹⁸ After 90 days, subjects using Meriva benefited from a more consistent control in skin discomforts and in the occurrence of skin breaks (from 100% to 63.6%). Individuals using Meriva also reported a greater reduction in oxidative stress, improvement in skin oxygenation and flux, and improvement of skin properties, such as oil and water content, hydration and elasticity.

Antidiabetic benefits

Type 2 diabetes causes the body to develop insulin resistance and, ultimately, results in elevated blood glucose levels that, if unchecked, can lead to severe health issues including kidney disease, vision impairment and heart disease.

Type 2 diabetes affects more than 30 million Americans, [according to](#) the CDC, underlining the need for solutions.



Creative approaches to branding and product positioning – podcast

Marc Brush, principal at the Bend consultancy, honed his market research bona fides running Nutrition Business Journal, but now he's helping shape the pursuit of personalized health care and nutrition.

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[†] Cox KHM et al. Nutrients. 2020 Jun 04. 12(6): 1678. doi: 10.3390/nu12061678
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Bioenhanced 95% curcuminoid turmeric extract (as BCM-95, from Arjuna Natural) showed significant antidiabetic and blood glucose-lowering effects in type 2 diabetic rats compared to regular turmeric extract.¹⁹ Going up against standard turmeric extract, BCM-95 was significantly better in reducing fasting blood glucose, reducing oral glucose tolerance tests at hours one and two, improving beta-cell function and insulin sensitivity, and reducing insulin resistance. Further, BCM-95 controlled blood glucose with less stress to the pancreas and with less insulin by improving sensitivity and reducing resistance compared to standard turmeric extract.

“[These] benefits are due to significantly higher levels of free curcumin in the pancreas with BCM-95,” explained Benny Anthony, Ph.D., joint managing director at Arjuna Natural. “The study provides the scientific rationale for use of BCM-95 in diabetic and prediabetic populations to control blood glucose, improve beta-cell function, insulin sensitivity and decrease insulin resistance.”

COVID-19 recovery

More than a year into the COVID-19 pandemic, medical professionals and health regulators are still looking for effective treatments for the virus that has taken the lives of more than 5 million people around the globe, according to the World Health Organization (WHO).

Dietary supplements—and curcumin, specifically—could answer the call for COVID-19 support. One 2021 clinical trial showed supplementation with 525 mg curcumin with 2.5 mg piperine twice daily could speed the recovery of patients with COVID-19.²⁰ Those in the curcumin group, compared to control, experienced early recovery from symptoms such as fever, cough, sore throat and breathlessness; less deterioration; better ability to maintain oxygen saturation; and better clinical outcomes. Other possible benefits of curcumin/piperine treatment include reduced duration of hospitalization in patients with moderate to severe symptoms, and fewer deaths.

Researchers attributed these effects to the anti-inflammatory and antithrombotic properties of curcumin.

Advancements in efficacy

Curcumin's potential to provide powerful health benefits to consumers may be contingent upon its ability to be absorbed and used by the body. Low bioavailability (attributed to water insolubility) and rapid metabolism to inactive metabolites are hurdles that stand in the way of curcumin's effectiveness.²¹

For this reason, some ingredient developers have worked hard to ensure the curcumin ingredients they supply will provide therapeutic effects to end users. These efforts are evidenced by a wealth of studies evaluating the bioavailability and efficacy of the ingredients.

In a single-dose, double-blind, two-way, crossover study of 12 volunteers, 225 mg curcuRouge (90 mg curcumin) demonstrated a C_{max} (peak serum concentration) and AUC_{0–8h} (concentration from administration to eight hours after) 5.4- and 3.4-fold higher, respectively, than those of another curcumin product (Theracurmin).²²

“Since native curcumin has extremely low bioavailability in humans and almost doesn’t show any dose-dependency, a comparative study with products of the market seems like an interesting way to assess bioavailability improvement,” Egoumenides said of the findings.

According to Rachela Mohr, manager for technical marketing and scientific management of bioingredients at WACKER, the bioavailability of curcumin can be boosted with the aid of gamma-cyclodextrin. She cited a 2018 study showing the curcuminoids of WACKER’s cyclodextrin complex CAVACURMIN were absorbed around 40 times more efficiently in the body than those of a standard curcumin extract.²³

Findings of a randomized, double-blind, 2021 study found that the metabolic bioavailability of tetrahydrocurcumin is also around 40 times higher following ingestion of CAVACURMIN, compared to a standard curcumin extract.²⁴

Tetrahydrocurcumin is a curcumin metabolite credited with additional health-promoting properties, Mohr explained. “Another finding of the long-term study is that, after only four weeks, daily ingestion of CAVACURMIN (approximately 340 mg curcumin each time) leads to a consistently high blood concentration of curcumin and tetrahydrocurcumin,” she said. “This effect can be measured after 12 weeks, as well.”

According to in vitro studies, Indena’s curcumin formulation with lecithin has shown a favorable effect on the natural biotransformation of curcuminoids by intestinal microbiota with a more efficient production of curcuminoids metabolites,²⁵ “opening new perspectives in investigation of curcuminoids metabolism and bioavailability,” Antonella Riva, product research manager, R&D, Indena S.p.A., said. “Research completed with Meriva continues to demonstrate its benefits as an antioxidant and for low chronic inflammation that are important to consumers.”

Pharmako Biotechnologies’ HydroCurc uses LipiSpense technology, which protects curcumin from hydrochloric acid during transit through the stomach to address curcumin’s poor bioavailability, limited biodistribution and poor gastrointestinal (GI) stability. In a 2019 study, researchers found HydroCurc consistently outperformed the standard C. longa extract, demonstrated drastic improvement on the solubility of curcuminoids, impacted the dissolution profile, and assisted in the permeation of curcumin across a CaCo2 cellular monolayer to show a 1.5-fold increase in the transport of curcumin across the CaCo2 monolayer.²⁶

A newly launched curcumin ingredient from OmniActive (Curcuwin Ultra+) is formulated using a specialized process to better protect and deliver more curcuminoids than standard curcumin, per unpublished research. In the randomized, double-blind, single-dose, crossover, pharmacokinetic study, Curcuwin Ultra+ demonstrated 40% faster absorption



of curcuminoids in the blood and 144 times greater relative absorption compared to standard curcumin. Add to that “an excellent organoleptic profile,” Deshanie Rai, vice president of global scientific and regulatory affairs at OmniActive, said, the ingredient is suitable across forms such as ready-to-drink (RTD) beverages and gummies.

Research continues to be a critical tool unlocking the potential of curcumin, allowing natural products manufacturers, marketers and consumers to seize the power of this super compound. ♦



Rachel Adams joined Informa's Health & Nutrition Network in 2013. Her career in the natural products industry started with a food and beverage focus before transitioning into her role as managing editor of Natural Products INSIDER, where she covered the dietary supplement industry. Adams left Informa Markets in 2019.

References

- 1 Vollono L et al. "Potential of Curcumin in Skin Disorders." *Nutrients*. 2019;11(9):2169.
- 2 Panahi Y et al. "Effect of Curcumin on Severity of Functional Dyspepsia: a Triple Blinded Clinical Trial." *Adv Exp Med Biol*. 2021;1308:119-126.
- 3 Lopresti AL et al. "Efficacy of a curcumin extract (Curcugen) on gastrointestinal symptoms and intestinal microbiota in adults with self-reported digestive complaints: a randomised, double-blind, placebo-controlled study." *BMC Complement Med Ther*. 2021;21(1):40.
- 4 Peterson CT et al. "Effects of Turmeric and Curcumin Dietary Supplementation on Human Gut Microbiota: A Double-Blind, Randomized, Placebo-Controlled Pilot Study." *J Evid Based Integr Med*. 2018;23:2515690X18790725.
- 5 Lorinczova TH et al. "Co-Administration of Iron and a Bioavailable Curcumin Supplement Increases Serum BDNF Levels in Healthy Adults." *Antioxidants (Basel)*. 2020;9(8):645.
- 6 Cox KH et al. "Investigation of the effects of solid lipid curcumin on cognition and mood in a healthy older population." *J Psychopharmacol*. 2015;29(5):642-651.
- 7 Cox KHM et al. "Further evidence of benefits to mood and working memory from lipidated curcumin [Longvida] in healthy older people: A 12-week, double-blind, placebo-controlled, partial replication study." *Nutrients*. 2020;12(6):1678.
- 8 Ngolab J et al. "Feasibility study for detection of retinal amyloid in clinical trials: The anti-amyloid treatment in asymptomatic AD (A4) trial." *Alzheimers Dement (Amst)*. 2021;13(1):e12199.
- 9 Tadokoro K et al. "Retinal amyloid imaging for screening AD." *J Alz Dis*. 2021;83(2):927-934.
- 10 Hillman A et al. "Ten Days of Curcumin Supplementation Attenuates Subjective Soreness and Maintains Muscular Power Following Plyometric Exercise." *J Diet Suppl*. 2021;1-15. DOI:10.1080/19390211.2021.1875101.
- 11 Mallard A et al. "Curcumin Improves Delayed Onset Muscle Soreness and Postexercise Lactate Accumulation." *J Diet Suppl*. 2021;18(5):531-542.
- 12 Schwarz NA et al. "A double-blinded, placebo-controlled trial evaluating the anti-inflammatory effects of Curcugen in an acute exercise mode." *Proceedings of the Seventeenth International Society of Sports Nutrition (ISSN) Conference and Expo. J Int Soc Sports Nutr*. 2020;17:59.
- 13 Delecroix B et al. "Curcumin and Piperine Supplementation and Recovery Following Exercise Induced Muscle Damage: A Randomized Controlled Trial." *J Sports Sci Med*. 2017;16:147-153.
- 14 Raj JP et al. "Effect of Turmacin supplementation on joint discomfort and functional outcome among healthy participants – A randomized placebo-controlled trial." *Complement Ther Med*. 2020;53:102522.
- 15 Kishimoto A et al. "Newly Developed Highly Bioavailable Curcumin Formulation, curcuRouge, Reduces Neutrophil/Lymphocyte Ratio in the Elderly: A Double-Blind, Placebo-Controlled Clinical Trial." *J Nutr Sci*. 2021;67(4):249-252.
- 16 Cherayil B. "Iron and immunity: immunological consequences of iron deficiency and overload." *Arch Immunol Ther Exp (Warsz)*. 2010;58(6):407-415.
- 17 Lorinczova HT et al. "Acute Administration of Bioavailable Curcumin Alongside Ferrous Sulphate Supplements Does Not Impair Iron Absorption in Healthy Adults in a Randomised Trial." *Nutrients*. 2021;13(7):2300.
- 18 Togni S et al. "Oral curcumin (Meriva) reduces symptoms and recurrence rates in subjects with atopic dermatitis." *Esper Dermatol*. 2019;21(2-4):42-46.
- 19 Sayeli VK and Shenoy AK. "Antidiabetic effect of bio-enhanced preparation of turmeric in streptozotocin-nicotinamide induced type 2 diabetic Wistar rats." *J Ayurveda Integr Med*. 2021;12(3):474-479.
- 20 Pawar K et al. "Oral Curcumin With Piperine as Adjuvant Therapy for the Treatment of COVID-19: A Randomized Clinical Trial." *Front Pharmacol*. 2021;12:669362.
- 21 Stohs S et al. "Highly Bioavailable Forms of Curcumin and Promising Avenues for Curcumin-Based Research and Application: A Review." *Molecules*. 2020;25(6):1397.
- 22 Sunagawa Y et al. "A novel amorphous preparation improved curcumin bioavailability in healthy volunteers: A single-dose, double-blind, two-way crossover study." *J Funct Foods*. 2021;81:104443.
- 23 Purpura M et al. "Analysis of different innovative formulations of curcumin for improved relative oral bioavailability in human subjects." *Eur J Nutr*. 2018;57:929-938.
- 24 Hundhammer C et al. "Enhanced metabolic bioavailability of tetrahydrocurcumin after oral supplementation of a γ-cyclodextrin curcumin complex." *J Funct Foods*. 2021;79:104410.
- 25 Bresciani L et al. "The Effect of Formulation of Curcuminoids on Their Metabolism by Human Colonic Microbiota." *Molecules*. 2020;25(4):940.
- 26 Briskey D et al. "Increased bioavailability of curcumin using a novel dispersion technology system (LipiSpere)." *Eur J Nutr*. 2019;58(5):2087-2097.



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Curcumin: Keeping abuzz with integrity, innovation

by Pete Croatto



Whenever an ingredient reaches “buzzworthy” status, a somber call for calm and the following of proper protocols echo in every Slack chat. Never mind that “these issues are becoming inherent to any active ingredient,” said Laure Egoumenides, product manager at Robertet. “But they are unfortunately emphasized when ingredients have such a high potential.”

Curcumin (*Curcuma longa*, derived from the turmeric plant) finds itself—literally and figuratively—under the microscope. Kristen Marshall, digital marketing manager at Verdure Sciences, explained that part of its popularity is the increase in demand for botanicals “since the onset of COVID, as consumers drive more toward natural and traditionally used supplements.”

The appeal of curcumin comes from an unsexy fact: It works. Sabinsa founder and president Muhammed Majeed, Ph.D., maintained curcumin has “delivered the promised health benefits to millions of people across the globe and earned their trust.”

According to Len Monheit, executive director of the Global Curcumin Association, that reputation dates back to ayurvedic traditions. Curcumin, he added, boasts mounds of science, especially with inflammatory markers,¹ and comes in an array of formats. Benny Antony, Ph.D., joint managing director at Arjuna Natural, said if curcumin is absorbed well, it can boost immunity and help with mood and cognitive support.²

“Curcumin supports various health outcomes, flowing beyond known lists,” Antony stated, adding gut health, oral health, skin care and athletic performance as relevant categories. “Curcumin boosts the antioxidant capacity of the body as it neutralizes free radicals on its own and also stimulates the body’s antioxidant enzyme.”³ This is a great development in the nutraceutical industry’s quest to find what he referred to as a “golden ingredient that could support and manage many more health conditions naturally.” The food and pharmaceutical industries, he suggested, have also taken notice.

Therein lies the rub. Any “golden ingredient” will attract its share of chuckleheads cutting corners in the never-ending quest to make a quick buck. “Adulteration is real, and the adulteration of turmeric is economically motivated,” Majeed pointed out.

Brandon Stolar, sales manager at Indena USA—makers of Meriva, a curcumin phytosome—contended the curcumin market remains solid with significant competition. “Each company is looking for advantages to stand out in the market. Features such as bioavailability or potency offer differentiation among many options and may be promoted positively or to make a product suspect in the marketplace,” he shared. “While competition is good to grow the market, it’s disreputable when companies take advantage of a growing market to adulterate products.”

On the positive side, competition has created a crop of curcumin ingredients with high standards beyond the lab. Product manufacturers can also choose an ingredient supplier that champions righteous social and business values while delivering real results to consumers. Ingredient integrity is a sure starting point, though.

Bioavailability and curcumin: An imperfect union

Rachela Mohr, manager for technical marketing and scientific management of bioingredients at WACKER, noted her company's concern with curcumin's bioavailability prompted the organization to focus on increasing it. Since the ingredient doesn't "readily dissolve" in water, she explained, the body eliminates up to 90% of curcumin taken orally.⁴

"Several approaches have been utilized to enhance bioavailability and bioactivity," Monheit stated. "This includes the addition of black pepper extract or other strategies to deliver more active. The challenge with bioavailability is on the one hand, it acknowledges this ingredient's challenge, and secondly, there are so many competing strategies and tools to measure bioavailability—both in vitro and in vivo. It's very difficult to get expert consensus on the best technology and approach and so we end up with an 'arms race' mentality."

Majeed and Antony insisted metabolism must be considered to accurately measure the bioavailability of curcumin.

"Bioavailability is to be considered comparing the absorption of free curcumin, which is responsible for the beneficial aspects of curcumin," Antony said. "The process of metabolism also releases curcumin metabolites, which are inactive." Arjuna's BCM-95, a 100% pure turmeric extract, evaluated free curcumin, but Antony explained a lot of companies either don't mention the number or opt for "an easier method to consider curcumin metabolites."

The results lead to the consumer getting shortchanged, Majeed held. "Many brands claim several tens of times enhanced bioavailability without a proportionate advantage in health effects, exposing the hollowness in their claims," he revealed. "Bioavailability is often measured with the wrong methodology. Incorrectly claimed enhanced bioavailability helps to market suboptimal amounts of curcumin. Intestinal effects of curcumin are often ignored, including their beneficial impact on microbiota.⁵ The biotransformation of curcumin into reductive metabolites is invariably glossed over."

Research, testing is everyone's responsibility

How does one navigate this particular pitfall?

"The key is to look at what ingredient has been substantiated and under what conditions and for what purpose/claim," Monheit detailed. "We've all got to be properly crediting and attributing the science to those companies, researchers and circumstances. If a finished brand does their own research, that's even better, as it will be on the exact in-bottle formulation."

WACKER did just that via studies showing that curcumin's bioavailability increases with the aid of gamma-cyclodextrin.

"A study published in 2018 showed that the curcuminoids of WACKER's cyclodextrin complex CAVACURMIN were absorbed around 40 times more efficiently in the body than those of a standard curcumin extract," Mohr said.⁶ "As part of a randomized double-blind study, it has now been shown that the metabolic bioavailability of tetrahydrocurcumin [a curcumin metabolite] is also around 40 times higher following ingestion of CAVACURMIN compared to a standard curcumin extract."⁷



“The challenge with bioavailability is on the one hand, it acknowledges this ingredient’s challenge, and secondly, there are so many competing strategies and tools to measure bioavailability—both in vitro and in vivo.”

Len Monheit, executive director of the Global Curcumin Association

Egoumenides said Robertet sources its turmeric extract—a clean hydroethanolic extract—from India. “Our partner, with over 13 years of work on curcumin and 50 clinical trials, brings an undeniable technical knowledge,” she stated. Two clinical trials on curcuRouge in cognitive health and immune health have already been conducted.

Majeed mentioned the term “bioavailability” has been misconstrued by both suppliers and marketing companies. “As a result, such products end up with low quantity of active curcuminoids in their dosages, with additional excipients, without sufficient science. Most importantly, the brands should assess the bioavailability from a functional point of view, and demonstrate if such bioavailability has improved any outcome of health effect.”

As is their wont, Majeed and Sabinsa back up the talk with scientific research on the company’s Curcumin C3 Complex (relevant studies are listed on the ingredient’s web page). “The effect of curcuminoids on the microbiome has been one area of interest for scientists at Sabinsa,” he said. “Sports nutrition is another area where the emphasis has been to safely manage post-workout inflammation.”⁸

Under pressure

Reputable curcumin ingredient manufacturers stand behind their products’ quality. “By owning our own fields and partnering with growers specializing in naturals all over the world, traceability, sustainability and transparency are more than just words for us—they are a fact,” Egoumenides asserted.

But manufacturers still need to fact check, as adulteration remains a major concern.

Elan Sudberg, CEO of Alkemist Labs—a fixture in testing botanical plants—identified “adulteration with undeclared synthetic material” as the trickiest quality problem with curcumin.

Monheit added the issue is—*what a shock*—not on FDA’s checklist. He recommended “a dual approach to testing using HPLC [high-performance liquid chromatography] and C [carbon]-14 testing.”

But Sudberg cautioned that may not be enough. Detecting synthetics “requires very specialized, insanely expensive testing equipment, which is why everyone we know sends curcumin to the University of Georgia Center for Applied Isotope Studies [CAIS] for radiocarbon testing to confirm it is derived from turmeric and not petrochemicals,” he said. Testing is more of a necessity as the perpetually strained supply chain has forced manufacturers to source material from unvetted vendors, Sudberg added.

Marshall agreed the COVID-fueled interest in botanicals has come with a price for vendors. “Increased demand has consequently impacted the supply of several botanicals, as suppliers try to keep up with the demand—especially with mandated lockdowns in growing countries like India and unpredictable and extreme weather patterns,” she shared. “Ongoing challenges with global logistics have increased



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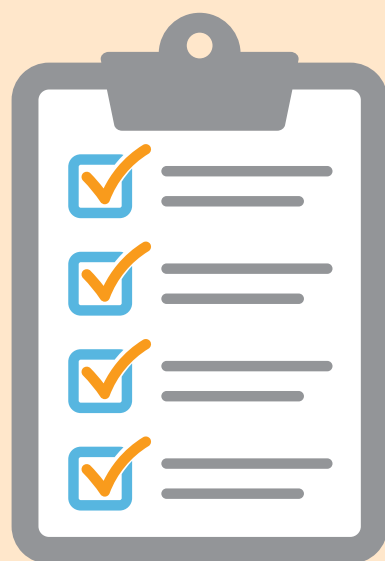
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Trends: Market pressures

both costs and lead times. As with many resources, limited growing seasons, increased demand, uncertain weather, global lockdowns, logistical delays and increased prices have all had a part in the new challenges our industry has faced throughout the pandemic.”

Ingredient suppliers and manufacturers must take an exhausting series of steps to ensure that it's business as usual.

Antony said Arjuna Natural works with its suppliers to share best practices and try to improve the supply chain, but that there are “more internal microprocesses and procedures that lead to the final success of a solid supply chain.” Among them, he listed:



Creating a supply chain emergency plan.



Building up inventory.



Conducting a supply chain vulnerability check.



Identifying backup suppliers.



Diversifying supply base.



Partnering with a logistics expert.



Adopting risk-evaluation tools.



Majeed contended Sabinsa not only has enough stock of raw materials on hand to ensure uninterrupted supplies, the company also has expanded its supply chain to help account for weather-related interruptions.

Numerous companies have made efforts to be better stewards of the land. Verdure Sciences (the maker of Longvida Optimized Curcumin) has explored new growing and harvesting techniques, as well as regenerative agricultural practices such as crop rotation and soil booster mixes. The company established a Verdugration initiative, which Marshall described as “practices of collaborating” with relevant parties “to integrate sustainable, traceable and certifiable practices.” Under that is Verdure Cares, which broadens its conscientious action to philanthropy, employee benefits and “sustainable partnerships for a long-lasting, holistic approach.”

Stolar stated Indena USA follows good agricultural and collection practices (GACP) in sourcing curcumin and has set local and global standards that align with the United Nations 2030 Agenda.

WACKER's bent is mutual compliance. “We comply with the OECD [Organisation for Economic Co-operation and Development] Guidelines for Multinational Enterprises, the ILO [International Labour Organization] core labor standards and the UN Guiding Principles on Business and Human Rights,” Mohr said. “To ensure that our suppliers meet our standards, we rely on various measures. For example,

WACKER has been a member of the Together for Sustainability (TfS) initiative to protect human rights in the supply chain since 2015. As part of TfS, suppliers are regularly audited, both by us and by other TfS companies. Best practices such as the United Nations Global Compact and the Responsible Care Global Charter are used.”

The underlying theme in the curcumin narrative is a tenet many people have embraced their whole lives—forming smart partnerships.

“The most important thing to do is to choose wisely, trusted partners and suppliers,” Egoumenides advised. “Of course it takes work and time for a supplier to set up a solid supply chain and develop high-quality ingredients, but also for manufacturers to identify and choose the most reliable solutions. A trustworthy ingredient must include sustainable and controlled sourcing, full traceability and a high quality control. It must integrate bioavailability issues and, last but not least, efficacy data.”

Monheit admitted curcumin has challenges. Not every supplier is superior, and the mindsets of commoditization and source redundancy need to be addressed. But the ingredient’s “buzziness” he said, is not going away. He predicted more industry and academic partnerships will help expand the science as “we get deeper into mechanistic substantiation.”

In short, skeptics need to find another target. “There’s a base of science associated with each highly invested supplier (the ones that have done the work),” Monheit stated, “and that needs to be respected—and attributed—in media and elsewhere.” ♦



Pete Croatto was an editor at VRM Inc. from 2003 to 2006 and ran Informa’s Supplement Perspectives blog from March 2011 to January 2015. His writing has appeared in many publications, including The New York Times, Columbia Journalism Review, GQ.com, and AARP the Magazine. His first book, “[From Hang Time to Prime Time: Business, Entertainment, and the Birth of the Modern-day NBA](#),” was released in December 2020.

References

- 1 Hewlings SJ and Kalman DS. “Curcumin: A Review of Its Effects on Human Health.” *Foods*. 2017;6(10):92.
- 2 Goozee K et al. “Examining the potential clinical value of curcumin in the prevention and diagnosis of Alzheimer’s disease.” *Br J Nutr*. 2016;115(3):449-465.
- 3 Hejazi J et al. “Effect of Curcumin Supplementation During Radiotherapy on Oxidative Status of Patients with Prostate Cancer: A Double Blinded, Randomized, Placebo-Controlled Study.” *Nutr Cancer*. 2016;68(1):77-85.
- 4 Lopresti AL. “The Problem of Curcumin and Its Bioavailability: Could Its Gastrointestinal Influence Contribute to Its Overall Health-Enhancing Effects?” *Adv Nutr*. 2018;9(1):41-50.
- 5 Peterson et al. “Effects of Turmeric and Curcumin Dietary Supplementation on Human Gut Microbiota: A Double-Blind, Randomized, Placebo-Controlled Pilot Study.” *J Evid Based Integr Med*. 2018;23:1-8.
- 6 Purpura M et al. “Analysis of different innovative formulations of curcumin for improved relative oral bioavailability in human subjects.” *Eur J Nutr*. 2018;57(3):929-938.
- 7 Hundhammer et al. “Enhanced metabolic bioavailability of tetrahydrocurcumin after oral supplementation of a γ -cyclodextrin curcumin complex.” *J Funct Foods*. 2021;79:104410.
- 8 Hillman AR et al. “Ten Days of Curcumin Supplementation Attenuates Subjective Soreness and Maintains Muscular Power Following Plyometric Exercise.” *J Diet Suppl*. 2021. DOI:10.1080/19390211.2021.1875101.



Anti-aging scientist IDs ingredient that extends life span and health span – podcast

Veteran anti-aging researcher Brian Kennedy, director of the Center for Healthy Longevity at the National University of Singapore, discusses alpha-ketoglutarate (LifeAKG) study results and the regulation of biomarkers.

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Innovation and versatility drive curcumin's future growth

by Joy Choquette



Aches and pains used to mean standing in front of the medicine cabinet searching for a bottle of painkillers. But in recent years, curcumin is a popular choice, with more individuals seeking out its golden goodness.

Curcumin isn't a newly discovered ingredient, though. Its roots grow deep: It's been used for centuries in traditional medicine. Curcumin has, however, experienced a resurgence of interest in recent years. More people are looking to this ingredient to potentially improve their health while hoping to reduce the effects of chronic pain, inflammation and other maladies.

Curcumin: A brief history

Curcumin has a long history in both Chinese and Indian traditional medicine practices. The *Curcuma longa* Linn plant—from which turmeric is obtained—is found in both China and India, as well as other areas of Southeast Asia. Distinctive with its bright yellowish-orange color, the 2019 scientific text “Curcumin for Neurological and Psychiatric Disorders” notes curcumin has been used to treat a variety of ailments, including rheumatism, anorexia, jaundice, diabetic wounds and hepatic disorders.

Gaining a lot of positive press in recent years, turmeric is a highly concentrated source of curcumin and one that many manufacturers have been studying and using in a variety of products. Technologies play an important role in helping curcumin to be used in novel ways.

Innovations in curcumin

Though turmeric is a leading botanical ingredient for joint health and is well-studied, said Abey Thomas, head of global marketing at Natural Remedies, it is not soluble in aqueous environments, such as the gastrointestinal (GI) tract. This, he noted, is due to curcumin's polarity and reduced dissolution rate. “As a result, it is poorly absorbed in the system,” Thomas noted. “In addition to the bioavailability issue, curcumin is typically extracted with harsh industrial solvents, a concern for many consumers in the age of clean label products.”

Despite the challenges, the company was committed to the plant's potential. “After years of research by our team of herbalists and scientists, we found that turmeric also contains a particular class of water-soluble polysaccharides that are potent, promising and bioactive, which led to the discovery of our patented extract Turmacin.”

Cheryl Myers, chief of scientific affairs and education at EuroPharma Inc., said curcumin is showing up in pharmacies and grocery stores in a variety of products. “For the past several years, there have been unique or novel inclusions of curcumin—or turmeric powder—used in foods, like breakfast cereals, and being offered as supplements in various forms, including chewables, powders and gummies,” she stated.

Regardless of the product format, Myers also stressed that the ability of curcumin to be readily absorbed in the body is key. “It is important to ascertain whether any of these delivery systems or foods are effective in terms of reducing inflammation, cell damage or joint pain,” she suggested.

Similar to the way that calcium is best taken with vitamin D3 for absorption,¹ other ingredients can be added to curcumin to improve its efficacy. Prafulla Kumar, an advisor for ProbioticSmart LLC, pointed to new research indicating the potential of metal complexes to improve curcumin bioavailability.² One of the company's products, TurmiZn-OSTM, combines curcumin, zinc and tetrahydrocurcumin. This stable complex provides an alternate to curcumin on its own.

As for the market's potential, Kumar maintained curcumin is experiencing steady growth, per [data](#) from Global Market Insights. "The curcumin market size exceeded US\$70 million in 2020 and is estimated to grow at a CAGR [compound annual growth rate] over 11% from 2021 to 2027," he shared.

Capsules, tablets, gummies and more

Capsules, and tablets, and gummies, and drinks ... oh my! The world of supplement varieties continues to grow. As noted in a Natural Products Insider article from the Council for Responsible Nutrition's (CRN) Brian Wommack, "[Supplements get personalized with new technology and delivery formats](#)," age seems to be the biggest influencer on which delivery method is preferable. While tablets and caplets remain the most popular among supplement fans as a whole, users in the 18- to 34-year-old demographic prefer gummies. Manufacturers who are aware of this and other trends are more easily able to respond to consumer demands.

EuroPharma offers a chewable curcumin supplement in its suite of CuraMed products, as well as the clinically studied BCM-95 ingredient. "Our curcumin is combined in a patented process with turmeric essential oil and has 700% better bioavailability than standard curcumin extracts, which are difficult for the body to absorb," Myers explained.³

Natural Remedies offers Non-GMO Project-certified Turmacin in traditional dosage forms like capsules and tablets. Thomas added the ingredient can also be formulated in functional food and beverages due to its self-affirmed GRAS (generally recognized as safe) status.

Dessert doesn't usually come to mind when thinking of immune support, but cookies are on the menu at ProbioticSmart. Kumar noted the company's patent-pending TurmiZn-OS curcumin-zinc complex empowers a wide range of nutraceuticals, foods and beverages, including an immune-positioned cookie dough product.

Curcumin used in novel combinations

The ability to use an ingredient in novel ways helps companies that want to grow with the times and spur consumer demand. Myers said EuroPharma's CuraMed Acute Pain Relief—a combination of BCM-95 curcumin and BOS-10 Boswellia [*serrata*—was shown to deliver up to 10 times more acetyl-11-keto-B-boswellic acid (AKBA) than unstandardized forms.⁴



How to market branded ingredients with integrity – podcast

Eric Anderson has been at the forefront of marketing evidence-based, IP-protected ingredients to quality-forward responsible brands, and shares his insights with Insider Senior Editor Todd Runestad.

naturalproductsinsider.com/ingredients/how-market-branded-ingredients-integrity-podcast

“In addition to the bioavailability issue, curcumin is typically extracted with harsh industrial solvents, a concern for many consumers in the age of clean label products.”

Abey Thomas, head of global marketing at Natural Remedies

The product employs an emulsified matrix of black sesame extract oil in a small and easy-to-swallow liquid gel. “Black sesame seed oil has been traditionally used in ayurvedic medicine to aid in carrying and dispersing fat-soluble ingredients,” Myers explained. “Plus, it is an anti-inflammatory in its own right, which makes it doubly valuable in this formula.”^{5,6}

A 2020 study examined the potential of the delivery mechanism.⁷ “Emulsifying curcumin and Boswellia in black sesame seed oil appeared to help disperse both fat-soluble herbs in the intestines and into the bloodstream even more quickly,” Myers reported.⁴

Putting it together

Beyond curcumin product and ingredient advances, scientific exploration has also multiplied. In his Insider article, “[Curcumin Sales Follow Science](#),” Todd Runestad noted, “In 2019 alone, according to PubMed, 1,924 curcumin studies were published, including 913 human clinical trials.” And as Kumar mentioned earlier, the forecast for the continued growth of curcumin in the marketplace is good.

Innovative delivery methods, new research regarding the best absorption methods, and curcumin’s long-standing history all lend themselves to its increased popularity. Perhaps soon more Americans will start reaching for curcumin products.

“I wish that consumers knew how good curcumin is for them,” Myers shared. “I tend to tell people that it doesn’t have serious side effects—only side benefits.” ✨



Content writing and journalism are high on the list of Joy Choquette’s interests. Writing professionally for the past 12 years, her work has appeared in national magazines, regional newspapers, and on lots and lots of websites. She specializes in health and wellness, business, and environmental topics. Learn more by visiting her [website](#) or find her on [LinkedIn](#).

References

- 1 Christakos S et al. “Vitamin D and intestinal calcium absorption.” *Mol Cell Endocrinol*. 2011;347(1-2):25-29.
- 2 Prasad S et al. “Metal–Curcumin Complexes in Therapeutics: An Approach to Enhance Pharmacological Effects of Curcumin.” *Int J Mol Sci*. 2021;22:7094.
- 3 Antony B et al. “A pilot cross-over study to evaluate human oral bioavailability of BCM-95 CG (Biocurcuma) a novel bioenhanced preparation of curcumin.” *Ind J Pharm Sci*. 2008;445-449.
- 4 Kizhakkeedath R et al. “Clinical Evaluation of an Herbal Formulation, Rhuief, in the Management of Knee Osteoarthritis.” *Osteoarthritis Cartilage*. 2011;19(S1):S145-S146.
- 5 Phitak T et al. “Chondroprotective and anti-inflammatory effects of sesamin.” *Phytochemistry*. 2012;80:77-88.
- 6 Haghighian MK et al. “Effects of sesame seed supplementation on inflammatory factors and oxidative stress biomarkers in patients with knee osteoarthritis.” *Acta Med Iran*. 2015;53(4):207-213.
- 7 Rudrappa GH et al. “Efficacy of high-dissolution turmeric-sesame formulation for pain relief in adult subjects with acute musculoskeletal pain compared to acetaminophen: A randomized controlled study.” *Medicine (Baltimore)*. 2020;99(28):e20373.

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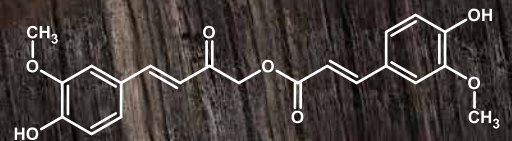
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Curcumin in 2021: Some answers, but questions remain

by Steve Myers



The curcumin supplement market has been awash in buzz on its potential in a range of health and performance situations, but the heightened absorption and bioavailability challenges have created a market full of different types—and qualities—of curcumin ingredients.

Responsible companies are investing in research and development (R&D) to improve these offerings, but for each published study seeming to provide an answer to which ingredient type best overcomes these challenges ... another study keeps the question open, creating discrepancies.

The past year is no different. Published studies added to the knowledge base and moved the market generally forward, but the competition and debate continue. If not for adulteration and the attention spans of modern consumers, this would be an entirely healthy situation where each tidbit of evidence and discovery, each one-upping in innovation of extraction and delivery technology, would boost the overall market.

One of the caveats of this “arms race,” as Global Curcumin Association Executive Director Len Monheit calls it, is that the latest evidence tends to be funded or directly studied by one of the competitors. While this demonstrates companies are putting money into research, it also can introduce the air of bias, or at least potential conflicts of interest.

The year started off with some new evidence of the real life benefits of curcumin supplementation. A January publication in the [Journal of Dietary Supplements](#) found curcumin is fit for sports and physical performance by decreasing inflammation and enhancing recovery.¹ This study was conducted by exercise and nutrition science professors from three U.S. universities and featured Jarrow Formula’s Curcumin 95 finished product, which uses Sabinsa’s Curcumin C3 Complex, a standardized extract (95% curcuminoids) of dried turmeric (*Curcuma longa* L.) rhizome. Specifically, the small study (n=22 healthy, recreationally trained males and females) found those taking curcumin for 10 days experienced decreased muscle soreness, compared to placebo, despite no significant overall differences in the inflammation marker creatine kinase (CK) between the groups. Additionally, curcumin users also saw less of a decline in vertical jump performance compared to those taking placebo.

The researchers noted varying results on curcumin and markers of both inflammation and muscle damage may be due to formulation, dosing and timing. They explained some studies use curcumin enhanced to improve bioavailability, while other studies, such as this one, use commercially available products not similarly enhanced. The dosages used for each of these types may need to be adjusted to see equivalent or comparable results.

Sabinsa markets a combination of C3 with its BioPerine black pepper (*Piper nigrum*) to improve bioavailability. The Jarrow Formula is among the top sellers on Amazon, but it doesn’t list piperine as an ingredient. A study published this year in [Frontiers in Pharmacology](#) showed supplementation with

the C3 and BioPerine combo twice daily in hospitalized COVID-19 patients also receiving conventional treatments substantially reduced mortality and morbidity.² Compared to control patients, the curcumin patients had earlier symptomatic recovery, less deterioration during their hospital stays and more consistent oxygen saturation above 94% without ventilation. While hospital stay lengths were similar between the mildly afflicted patients in both groups, stays were significantly less in moderate and severe patients on curcumin, and fewer of these patients required ventilation compared to their control group counterparts. Thus, in addition to the primary benefits, the researchers concluded curcumin supplementation in COVID patients could reduce treatment costs.

Arjuna Natural Pvt Ltd. also offers a standard turmeric extract (STE) with 95% curcuminoids, but this BCM-95/CURCUGREEN brand ingredient also features turmeric essential oil enriched to 45% ar-turmerone, an aromatic terpene from the rhizome. A pair of rodent studies published in 2021 shed some light on the potential benefits of this enhanced extract.

As outlined in an April 2021 publication in [Antioxidants](#) journal, a team of neuroscience hospital and university researchers in Michigan found BCM-95 reduced inflammation and degeneration in several organs (spleen, kidney, liver and lungs) and also decreased aspects of Alzheimer's disease pathology such as organ amyloid deposition, brain tissue phosphorylated tau (pTau), pro-inflammatory markers, cell loss and limited anti-inflammatory marker declines.³

On BCM-95 specifically, they noted it completely dissolved in water, whereas natural curcuminoids showed precipitation, the opposite of dissolution. Further, they deemed BCM-95 as 700 times more potent than natural turmeric and noted it can stay in the circulation system longer—eight to 10 hours, compared to two to four hours for natural curcuminoids. When looking at how much free curcuminoids were available in different tissues, they found 200 to 300 times more free curcuminoids from BCM-95 reached the blood, brain and other organs compared to natural turmeric curcuminoids.



Integrating botanicals for wellness – podcast

Cheryl Myers, chief of scientific affairs and education at EuroPharma, discusses a “smorgasbord” approach to wellness, which integrates a host of botanical ingredients that offer an array of potential benefits.

naturalproductsinsider.com/herbs-botanicals/integrating-botanicals-wellness-podcast



A pair of college researchers from India, where turmeric and curcumin are grown and have long been a part of the traditional medicine, also pitted BCM-95 against natural curcuminoids in rodents, demonstrating antidiabetic and blood-glucose-lowering benefits for the enhanced extract.⁴ Published in the [Journal of Ayurveda and Integrative Medicine](#), the study also credited BCM-95's ability to increase free curcuminoids in blood and organs, compared to regular turmeric extract.

While these individual studies comparing a single extract to a basic turmeric extract provide some level of confidence in improved bioavailability and benefits to specific areas of health, studies comparing several specialty extracts and finished products may provide a more comprehensive view of the range of bioavailability in the marketplace.

In April 2021, an online article ahead of print publication in the *Journal of Nutrition* looked at the pharmacokinetics of various forms of turmeric at the doses: STE with 95% curcuminoids (from Naturex, at 1,500 mg); liquid micellar curcumin containing 6% curcuminoids (NovaSOL from Aquanova AG, at 1,000 mg); a combo of piperine and 95% curcuminoid extract (Curcumin C3 Complex from Sabinsa, at 1,515 mg); a phytosome formulation of turmeric extract (18% to 22% curcuminoids), phosphatidylcholine (PC) and microcrystalline cellulose (Meriva from Indena, at 1,000 mg); and dried colloidal suspension of STE (30% curcuminoids), *Quillaja saponaria* extract, sunflower oil and acacia gum (Turmipure GOLD from Naturex, at 300 mg).

The amounts reflected the recommended doses, and ingredients were researched for increased bioavailability. The researchers explained, "Consumers will be able to make informed decisions regarding which turmeric formulation to choose based on their real curcuminoid absorption capacity." The curcuminoid content of each was verified using ultra-high-pressure liquid chromatography (UHPLC)/tandem mass spectrometry (MS/MS).

In the randomized, open-labeled, crossover design, 30 healthy men and women sequentially consumed single doses of these and had 11 blood samples each drawn over a 24-hour period, and researchers assessed the dose-normalized area under the curve (AUC) of total plasma curcuminoids. They found Turmipure GOLD was the most bioavailable low-dose extract of the bunch, providing "high unconjugated and conjugated curcuminoid absorption, with significant beneficial differences when compared with the high dose of standard extract."



It is important to note that the researchers were from Naturex, and the focus of the study was to compare Turmipure to market competitors. This isn't to suggest inaccurate results, only to highlight a potential conflict of interest. The study made it to [print in July 2021](#).⁵

Another comparison surfaced in 2021, this time an analysis of retail-purchased curcumin finished products. In this case, NOW Foods purchased curcumin/turmeric products from Amazon in June 2021 and blind-tested for potency, heavy metals, labeling accuracy and potential addition of synthetic curcumin. Of the 23 purchased brand products and two NOW products analyzed, only one clearly failed potency testing and four others tested very low, but without any specific label claim.

The labeling of products was an issue, with many saying "Turmeric Curcumin 1,650 mg" on the front of the label, but listing breakdown amounts on the back or side from various sources such as turmeric root, turmeric extract and piperine. This effectively meant the 1,650 mg was not entirely curcumin as indicated on the front. "This can be perceived as deceptive since many customers do not know the difference between turmeric, turmeric extract, curcumin extract, and standardized 95% extract," said Dan Richard, NOW's vice president of global sales and marketing.

NOW also tested the products for heavy metals, including arsenic, cadmium, lead and mercury. On average, total heavy metals were 525% higher in the other purchased brands than in the two NOW products, with only one product testing lower than NOW's. The company noted two products that turned out to be B'Leaf Nature and Eagle brands had more than 20 times higher heavy metals than NOW's products, and they also had lead levels above California's Prop 65 limits. Two others, Farm Haven and BioEmblem brands, had cadmium levels above 100 ppb (parts per billion).

Lastly, samples of the products were sent to the University of Georgia's Center for Applied Isotope Studies for independent radiocarbon testing to look for possible adulteration, a problem highlighted by the American Botanical Council (ABC). Of the 23 unfamiliar brands, evidence of spiking with "fossil fuel

derived organic carbon” was present in four samples that turned out to be from Vitpro, Me First Living, Eagle, and Primal Harvest. Further, two products labeled as vegetarian capsules— Bioganix and Nutriflair brands—tested positive for animal gelatin.

These results indicate some consumers are getting a raw deal when buying curcumin/turmeric supplements. NOW said Amazon is taking first steps to address these ongoing, egregious problems with sellers on its platform, but more needs to be done. “Amazon has consistently not given NOW any feedback on these testing results and remaining problems,” Richard said. “Amazon is moving forward with their stricter requirements for sports and diet products, as well as ISO17025 Certificates of Analysis [CoAs].” NOW further reported it was not aware that any product it tested has been reformulated or relabeled.

“It takes a lot to move mountains like curcumin,” Richard said. “Some quality producers of curcumin have told their story of absorption, naturalness, heavy metals, potency and overall quality for years, and yet the disparity in products remains. So, we didn’t expect to move mountains with our testing and reporting, though it would be nice to see some visible changes.”

The testing method and results were made public, but they are not published in a peer-reviewed journal as of press time.

See [pages 6](#) and [16](#) for more information on the latest curcumin efficacy and bioavailability research and technological developments. The overall takeaway from 2021 curcumin-related news and developments is this is a competitive market with bioavailability challenges, but each piece of research and innovative technology can further the knowledge and quality of the market. However, this new knowledge needs to be considered in the context of the research scope, testing methods and material parameters, including dosage and timing specifications, not to mention the competition influence. ✨

References

1. Hillman A et al. “Ten Days of Curcumin Supplementation Attenuates Subjective Soreness and Maintains Muscular Power Following Plyometric Exercise.” J Diet Suppl. 2021;1-15.
2. Pawar K et al. “Oral Curcumin With Piperine as Adjuvant Therapy for the Treatment of COVID-19: A Randomized Clinical Trial.” Front Pharmacol. 2021;12:669362.
3. Manna J et al. “Curcugreen Treatment Prevented Splenomegaly and Other Peripheral Organ Abnormalities in 3xTg and 5xFAD Mouse Models of Alzheimer’s Disease.” Antiox. 2021;10(6):899.
4. Sayeli VK and Shenoy AK. “Antidiabetic effect of bio-enhanced preparation of turmeric in streptozotocin-nicotinamide induced type 2 diabetic Wistar rats.” J Ayurveda Integr Med. 2021;12(3):474-479.
5. Fança-Berthon P et al. “Pharmacokinetics of a Single Dose of Turmeric Curcuminoids Depends on Formulation: Results of a Human Crossover Study.” J Nutr. 2021;151(7):1802-1816.

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